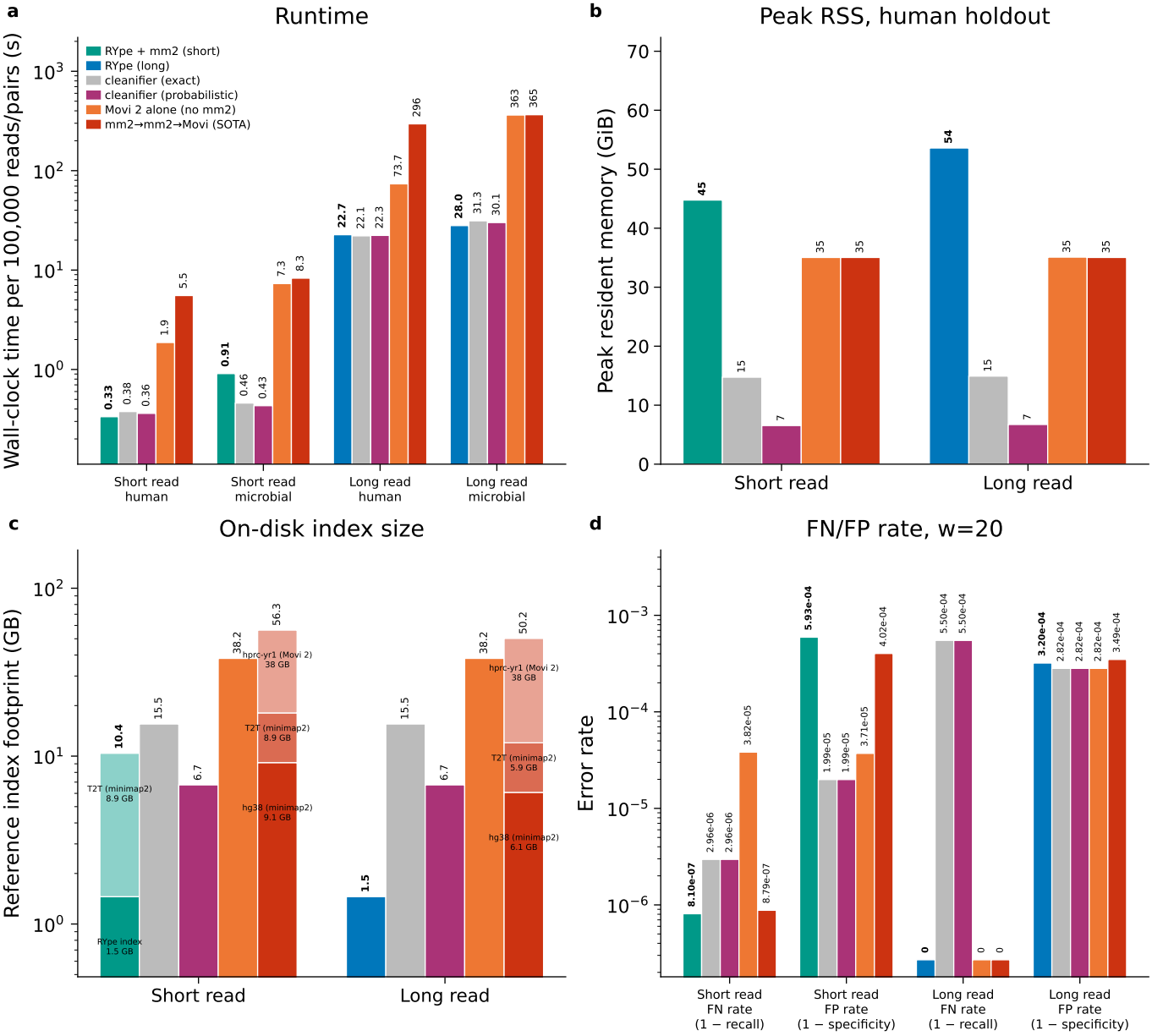
a
Runtime
RYpe + mm2 (short)
RYpe (long)
cleanifier (exact)
cleanifier (probabilistic)
Movi 2 alone (no mm2)
mm2→mm2→Movi (SOTA)
Wall-clock time per 100,000 reads/pairs (s)
0.33
0.38
0.36
1.9
5.5
0.91
0.46
0.43
7.3
8.3
22.7
22.1
22.3
73.7
296
28.0
31.3
30.1
363
365
Short read human
Short read microbial
Long read human
Long read microbial
b
Peak RSS, human holdout
Peak resident memory (GiB)
45
15
7
35
35
54
15
7
35
35
Short read
Long read
c
On-disk index size
Reference index footprint (GB)
10.4
15.5
6.7
38.2
56.3
T2T (minimap2) 8.9 GB
hprc-yr1 (Movi 2) 38 GB
T2T (minimap2) 8.9 GB
hg38 (minimap2) 9.1 GB
RYpe index 1.5 GB
1.5
15.5
6.7
38.2
50.2
hprc-yr1 (Movi 2) 38 GB
T2T (minimap2) 5.9 GB
hg38 (minimap2) 6.1 GB
Short read
Long read
d
FN/FP rate, w=20
Error rate
8.10e-07
2.96e-06
2.96e-06
3.82e-05
8.79e-07
5.93e-04
1.99e-05
1.99e-05
3.71e-05
4.02e-04
0
5.50e-04
5.50e-04
0
0
3.20e-04
2.82e-04
2.82e-04
2.82e-04
3.49e-04
Short read FN rate (1 − recall)
Short read FP rate (1 − specificity)
Long read FN rate (1 − recall)
Long read FP rate (1 − specificity)